Bangladesh University of Engineering and Technology
Department of Computer Science and Engineering

Course: CSE 206
Digital Logic Design Sessional

Experiment No. 7
Design Using Multiplexers

Design and implement the following problems:

1. Design and implement each of the two following functions using one 4x1 Multiplexer (IC – 74153) and necessary gates.
   - $f(a, b, c, d) = \sum(0, 1, 3, 5, 9, 10, 14)$
   - $f(a, b, c, d) = \prod(0, 1, 5, 9, 10, 12, 15)$
2. Design a 16x1 Multiplexer using 4x1 Multiplexer (IC – 74153) only. Using your designed Multiplexer, implement the function :
   $f(a, b, c, d) = \sum(0, 2, 3, 6, 10, 11, 13)$

Lab Sheet : For each of the problems, your lab sheet should cover

- Problem definition
- Truth table and minimized equation with minimization steps (if applicable)
- Circuit diagram with pin number.
- Required instruments for implementation.

Answer the following questions:

1. Design a 4 bit Binary to Gray code Converter using 4x1 Multiplexer and necessary basic gates. Show all the necessary steps of your implementation.
2. Design a excess-3 to binary code converter using 8x1 Multiplexer (IC-74151) and necessary basic gates. Show all the necessary steps of your implementation.